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The prokaryotic community of a flow regulated lagoon

Fabio De Pascale[1], Greta Battaggia[1], Irene Gregori[1,2], Riccardo Schiavon[3], Elisa Camatti[4], Florencia Mohamed[1], Chiara Facca[5] and Alessandro Vezzi[1]*

[1]Department of Biology, University of Padua, Padua, Italy, [2]National Biodiversity Future Center (NBFC), Palermo, Italy, [3]BMR Genomics, Padua, Italy, [4]Institute of Marine Science, National Research Council (CNR ISMAR), Venice, Italy, [5]Department of Environmental Sciences, Informatics and Statistics, Ca' Foscari University of Venice, Venice, Italy

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*CORRESPONDENCE
Alessandro Vezzi
alessandro.vezzi@unipd.it

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Microorganisms play a crucial role in all the environments they inhabit, including coastal areas. These ecosystems are facing increasing pressures due to climate change, rising sea levels, and mitigation measures. One such measure in the Venice lagoon is the MOSE system, a large-scale infrastructure designed to protect the city of Venice from tidal events by regulating water inflow from the sea. To understand the ecological consequences on microorganisms of this regulated seawater exchange, the present study analyzes benthic and pelagic prokaryotic communities of the Venice Lagoon before the systematic activation of the MOSE barriers. Water and sediment sampling was conducted every three months at five stations over a thirty-month period, and microbial communities were investigated using 16S rRNA metabarcoding. The samples were analyzed to describe microbial abundances and variation across different sites and seasons. The results revealed a significant difference between benthic and pelagic communities, although a considerable amount of species were shared among them. While pelagic communities showed strong seasonal variations, benthic communities were more temporally stable, but exhibited strong variability across sites. These results provide a detailed characterization of the prokaryotic communities associated with surface water and sediment, offering a valuable baseline for future studies assessing the impact of highly regulated coastal environments.

KEYWORDS

benthic, lagoons, MOSE system, pelagic, prokaryotic community, Venice

Introduction

The whole biosphere is inhabited by microbes. They are found in the most diverse habitats, from the highest mountain peaks to the depths of the ocean. Microbes are not only ubiquitous, but also play a fundamental role in all the ecosystems they inhabit. Since 70% of the Earth's surface is covered by oceans, it is clear that marine microbes are essential for life by performing key functions that support planetary health, such as nutrient cycling, climate modulation and water filtration (Sessitsch et al., 2023). While oceans and seas cover most of the planet, coasts represent a tiny portion of the planet and nonetheless host many living beings including humans - nearly half of the global population live within 100 km from the sea (Sea change in coastal science, 2020). Coasts represent the interface between the terrestrial and aquatic realms, boasting a wide range of different habitats, being home to some of the most biologically and geochemically active systems on Earth (U.S. DOE, 2017) and providing a wide range of ecosystem services (Ward et al., 2020). Lagoons are a common landform of coastal transitional environments found at the estuaries of many rivers all along the borders of most continents (Kjerfve and Magill, 1989).

Lagoons are complex and dynamic systems that are strongly influenced by physical and chemical variables, such as current and tidal flows, marine sediment and water imports, and terrestrial inputs from rivers and groundwater (Kjerfve and Magill, 1989). The semi-diurnal mixing of fresh and sea waters causes high physical and chemical variability in both space and time within these environments (Pérez-Ruzafa et al., 2005; Gačić et al., 2004). Also, relatively closed lagoon ecosystems are extremely exposed to nutrient input from the surrounding area, including anthropogenic pollution (Newton et al., 2014; Zonta et al., 2018). In this respect, the Venice lagoon, one of the largest transitional coastal ecosystems in the Mediterranean and a UNESCO World Heritage site (Madricardo et al., 2019), represents an invaluable study area for understanding how the biotic components adapt to the various and constantly changing environmental conditions, and how the different ecosystem components influence each other.

The Venice lagoon consists of different environments, such as salt marshes, an intricate network of channels, fish farms, natural and artificial islands. Today its total surface is 550 km^2, of which 390 km^2 of open lagoon including 40 km^2 of tidal channels, 70 km^2 of salt marshes and 90 km^2 of fish farms (Madricardo et al., 2019). Only 5% of the lagoon has a depth greater than 5 m, and 75% is less than 2 m deep resulting in an average depth of 1.2 m that imply a tight coupling between water and sediments (Molinaroli et al., 2007). It is subjected to a multitude of natural and anthropogenic stressors, including sea level rise and subsidence, the presence of the Porto Marghera chemical industrial area, and the massive tourist load on the Venice city (Solidoro et al., 2010). Lido, Malamocco, and Chioggia are the three inlets that currently connect the lagoon with the open sea (from North to South) with length around 2.5 km each, mean depth 10, 16, and 8 m, respectively, and width from 0.5 to 1 km (Umgiesser et al., 2014). The lagoon hydrodynamic and water exchange at the inlets are mainly influenced by the tide (Gačić et al., 2002), with the mean tidal range oscillating between 0.5 m during neap tide and 1 m during spring tide. The water residence time varies over the lagoon and according to weather conditions, but average values are on the order of a few days in areas close to the inlets and up to 40 days in the inner and more enclosed areas (Cucco et al., 2009). Based on hydrodynamic studies, the Venice lagoon can also be divided into four areas called sub-basin: the far-northern (NBn), the northern central (NBc), the central (CB) and the Southern (SB) (Solidoro et al., 2004).

Since its formation around 6000 years ago, it has undergone constant natural and anthropogenic changes, and substantial efforts have been made to stabilize and preserve its morphological and ecological features. As early as 1501, in fact, a specific water authority called *Magistrato alle Acque* was established by the Venetian Republic to manage the lagoon (Bevilacqua, 1998). The same authority operates nowadays to regulate interventions and activities in the lagoon.

The latest massive human intervention on this ecosystem has been the design and construction of a system of mobile barriers (MOSE - MOdulo Sperimentale Elettromeccanico) at the inlets in order to abate the maximum tide level within the Venice lagoon and thus protecting the historical city and the environment from floods (Trincardi et al., 2016) (Institute of Marine Sciences, CNR-ISMAR, Venice, Italy et al., 2020). MOSE, which became operational in October 2020, might substantially affect the lagoon hydrodynamics and, consequently, sediment transport and balance (Leoni et al., 2022).

Considering its relevance, many studies have been focused on the Venice lagoon, both from a chemical-physical and biotic point of view (see (Solidoro et al., 2010) for a comprehensive overview). However, few researches have addressed the characterization of the structure and dynamics of prokaryotic communities in this environment. Yet, prokaryotes play an important role in the lagoons functioning and productivity, since they mineralize organic matter, cycle biogeochemically relevant elements, degrade pollutants, are primary producers and transfer part of the energy to higher trophic levels (Schubert and Telesh, 2017; Trombetta et al., 2022).

Two early works investigated the bacterial community structure and diversity of the Venice lagoon through the sequencing of 16S rRNA gene clone libraries. A first study characterized bacterial and archaeal communities in sediment samples collected in May 2005 in the different sub-basins of the lagoon (Borin et al., 2009). The second research focused on bacterioplankton diversity and community composition within the southern basin of the lagoon, and the corresponding inlet, in two different seasons (Simonato et al., 2010). More recently, thanks to the advent of next-generation sequencing (NGS) technologies and their unprecedented high-throughput, two studies have increased our knowledge of the structure, dynamics, and genetic potential of microbial communities in the Venice Lagoon. In 2017, Quero and colleagues applied DNA metabarcoding to study the diversity of planktonic and benthic bacterial assemblages over a one-year period in a transect between the inner lagoon and the Gulf of Venice (Quero et al., 2017). A following study provided additional insights into the microbes living in the surface sediments of the Venice Lagoon in terms of their spatial and temporal distribution and dynamics (Banchi et al., 2021). Moreover, this work employed the metagenomic approach, thus shedding light for the first time on the potential functions and metabolism of microbes in the lagoon sediment.

In the present study, we seek to deepen our knowledge on the diversity and dynamics of prokaryotic communities inhabiting the Venice lagoon. To this end, we investigate the seasonal structure of planktonic and benthic prokaryotes over a two-and-a-half year time span through 16S rRNA gene metabarcoding. Furthermore, we monitored five different sampling sites to obtain insights on the microbial biodiversity existing in each of the hydrological sub-basins of the lagoon. These data will serve as a solid baseline for tracking the resilience and potential evolution of the lagoon's prokaryotic communities, especially now that the system of mobile barriers has become fully operational.

Materials and methods

Site description and sampling strategy

Surface water and sediments were collected at neap tide seasonally from November 2018 until May 2021 at five different sites within the Venice lagoon, Italy (Figure 1): Bocca di Chioggia (BCHIO), Bocca di Lido (BLIDO), Marghera (MARGH), Palude della Rosa (PROSA) and San Giuliano (SGIUL). The only exception was Bocca di Chioggia where sediments were collected from February 2019 and water samples from May 2019. The stations cover the four sub-basins into which the lagoon is divided from a hydrodynamic point of view (Solidoro et al., 2004). Moreover, four out of five stations are sites of the Italian Long-Term Ecological Research Network (LTER-IT) since 2007 (Mirtl et al., 2018) (Acri et al., 2020) (Supplementary Table 1). The sites had different depths ranging from 1 to 12.6 m and different sediment textures (see Supplementary Table 4). The stations were also selected because they represent the different extent of anthropogenic impacts on the lagoon. The Marghera site, being inside the petrochemical pole of Porto Marghera, is heavily affected by metal and organic contamination (Picone et al., 2016). The San Giuliano site is influenced by the freshwater inputs of the Osellino River which conveys nutrients and pollutants into the lagoon (Sfriso et al., 2008). The Bocca di Chioggia site, close to the southern lagoon inlet, as well as the Bocca di Lido one, in the lagoon's northernmost inlet, are under the direct influence of Adriatic coastal waters. Lastly, the Palude della Rosa site is located in a lightly contaminated mudflat area, representing a typical lagoon environment (Picone et al., 2016).

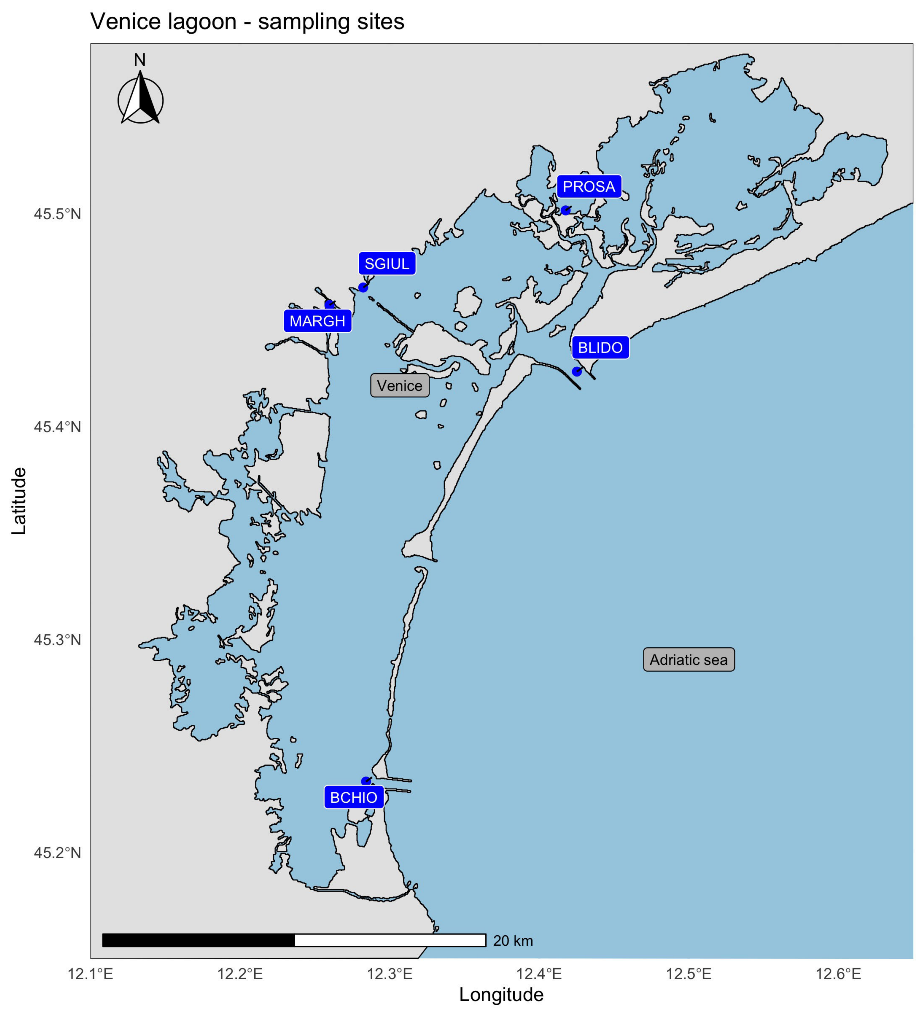

FIGURE 1
Map of the Venice Lagoon showing the five sampling sites: Bocca di Chioggia (BCHIO), Bocca di Lido (BLIDO), Marghera (MARGH), Palude della Rosa (PROSA) and San Giuliano (SGIUL).

Water samples were collected using 10 L flasks sterilized with 10% HCl and cleaned with deionized water afterwards. Flasks were immersed by hand at a maximum depth of 0.5 m below the surface, and filtered on-site. The filtration procedure was adapted from the *Tara* Oceans expedition (Pesant et al., 2015). In brief, water samples were flowed through three different pore size filters arranged in series: 20 µm (47-mm-diameter nylon mesh, Merck Millipore), 3 µm (142-mm-diameter polycarbonate membrane, Merck Millipore) and 0.22 µm (142-mm-diameter polyethersulfone Express Plus membrane, Merck Millipore). This size fractionation approach enables free-living prokaryotes to be captured in the 0.22-3 µm fraction, while particle-attached prokaryotes are retained in larger pore size fractions (3–20 µm and >20 µm) (Pascoal et al., 2023). Filters were immediately frozen on dry ice and stored at -80°C.

Sediment samples were collected with a Van Veen grab (0.045 m^2) at three different locations in a thirty-meter range around the GPS coordinates of each station (Supplementary Table 1). Depending on the sediment texture, the grab retrieved layers between 2 and 15 cm thick. To minimize potential inconsistencies in subsequent analyses, samples were collected with a spatula from the uppermost 1 cm of the recovered material. Approximately 30 mL of sediment was transferred into 50-mL Falcon tubes, immediately frozen on dry ice, and stored at −80°C.

Environmental parameters measurement

At all stations, temperature, salinity and percentage of dissolved oxygen of the surface water samples were measured using a CTD probe (Seabird 911, Sea-Bird Electronics, Washington, USA) (Supplementary Table 2). Transparency was measured with a Secchi disc. The concentration of dissolved inorganic nutrients including nitrogen as ammonium ($N\text{-}NH_4$), nitrites ($N\text{-}NO_2$) and nitrates ($N\text{-}NO_3$), phosphorous as orthophosphate ($P\text{-}PO_4$), and silicon as orthosilicate ($Si\text{-}SiO_4$) was determined with a Systea EasyChem Plus analyzer (Systea Analytical technologies, Anagni, Italy), in accordance with the colorimetric methods indicated by Grasshoff et al. (Grasshoff et al., 1985). Chlorophyll-*a* concentration was determined spectro-fluorometrically as described by Bernardi Aubry and colleagues (Bernardi Aubry et al., 2013). The number of photoautotrophic and heterotrophic prokaryotes was determined by flow cytometry and kindly provided by Mauro Celussi of OGS in Trieste, Italy (Supplementary Table 3). A detailed description of how the data were obtained can be found in Manna et al., 2019. Temperature, salinity, percentage of dissolved oxygen and pH of the water overlying the sediment samples were determined with the multiparametric probe HI98128 (Hanna instruments, Villafranca Padovana, Italy). Grain size composition was determined with a Mastersizer 3000 particle size analyzer (Malvern Panalytical, Malvern, United Kingdom) after passing each sample through a 1-mm sieve. Data were expressed as percentages of sand, silt and clay, following the Udden-Wentworth's classification to define the limits between the three size groups (Wentworth, 1922). Finally, sediments were described using Shepard's classification (Shepard, 1954) (Supplementary Table 4).

DNA extraction from filters

DNA was extracted from each 0.22 µm filter in order to primarily target free-living prokaryotes. Filters were cut into six pieces, and two were used independently for DNA extraction using the AllPrep DNA/RNA/miRNA Universal kit (Qiagen, Hilden, Germany) following the protocol for the simultaneous purification of nucleic acids from cells with slight modifications. Specifically, filter pieces were first incubated in lysozyme solution (0.25 mg/ml in TE) for 5 minutes at 37°C, without agitation (McCarthy et al., 2015). They were then transferred into 2-ml centrifuge tubes containing 600 µl of RLT lysis buffer supplemented with β-mercaptoethanol (0.140 M final concentration) were added to each tube. Tubes were incubated at room temperature for 90 minutes while rotating in a rotisserie. After the incubation, tubes were vortexed for 10 min at maximum speed and 4°C, and centrifuged for 1 minute at 10,000 *g*, with the filter pieces pressed under the cap to recover as much lysate as possible. The filter fragments were then removed, and lysates were pipetted at least five times through 1000-µl tips before transfer to the AllPrep DNA Mini spin columns. From this step onward, the manufacturer's protocol was followed.

DNA extraction from sediments

DNA was extracted from sediment samples using the RNeasy PowerSoil Total RNA kit in combination with the RNeasy PowerSoil DNA elution kit (Qiagen, Hilden, Germany), following the manufacturer's protocol. In order to minimize possible point variations at individual sampling sites, the sediments collected at the three different locations of each station were weighed and mixed evenly, and then used to perform two independent DNA extractions.

DNA quantification and PCR amplification

The concentration and purity of extracted DNA were assessed using a NanoDrop 2000 spectrophotometer and a Qubit 2.0 fluorometer (Thermo Fisher Scientific, Waltham, USA). Amplicon libraries were generated by PCR amplification of the V4 hypervariable region of 16S rRNA genes using the primer pair 515F-Y (5′-GTGYCAGCMGCCGCGGTAA-3′) and 806R (5′-GGACTACNVGGGTWTCTAAT-3′) (Apprill et al., 2015). PCR reactions were performed in 25-µl containing: 5 ng of template DNA, 0.5 U of Phusion High-Fidelity DNA polymerase (Thermo Fisher Scientific, Waltham, USA), 1X Phusion HF buffer, 0.5 µM of each primer and 200 µM of each dNTP. Thermal cycling conditions were: 98°C for 4 min; 25 cycles of 98°C for 20 s, 57°C for 30 s, 72°C for 30 s; 72°C for 5 min. Each sample was amplified in duplicate to reduce potential intra-sample variation. PCR products were checked on 1.5% agarose gels, pooled and purified using 0.6X volumes of AMPure XP beads (Beckman Coulter, Brea, CA, USA).

Library preparation and sequencing

Pooled PCR products were indexed and normalized following the "16S Metagenomic Sequencing Library Preparation" protocol (Illumina, San Diego, CA, USA), with the following modifications: 1) 1 U of Phusion High-Fidelity DNA polymerase and 1X Phusion HF buffer (Thermo Fisher Scientific, Waltham, USA) were used per reaction; 2) each denaturation step was performed at 98°C; 3) PCR amplicons were normalized using the SequalPrep Normalization Plate kit (Thermo Fisher Scientific, Waltham, USA). Amplicon libraries were pooled in equimolar amounts, purified using 0.8X volumes of AMPure XP beads (Beckman Coulter, Brea, CA, USA), and sequenced across three Illumina MiSeq runs in paired-end 2x300 format (Illumina, San Diego, CA, USA). The 16S amplicon sequences generated in this study can be found in the Sequence Reads Archive (SRA) at NCBI with the BioProject accession number PRJNA1096706.

Bioinformatic analyses

Raw reads were initially processed with Cutadapt (v. 2.1) (Martin, 2011) to remove primer sequences. All further analyses were conducted in R (v 4.0.1.). Initially, paired-end reads were trimmed, denoised and merged with the "DADA2" package to obtain the amplicon sequence variants for each sample (Callahan et al., 2016a, 2017). Reads were quality trimmed depending on the run; forward reads were trimmed at 220, 240 and 250 bases and reverse reads at 150 and 180 bases, with a maximum expected error rate of 2 (Supplementary Table 5 for details). Paired-end reads were merged with a minimum overlap of 10 bp and chimeras were removed using the "*consensus*" method. Taxonomy annotation was performed in DADA2 using the assignTaxonomy() function with SILVA v.138, followed by addSpecies function for species-level assignment (Quast et al., 2013; Yilmaz et al., 2014; Glöckner et al., 2017). Sequences that could not be assigned were labeled as NA.

Subsequent analyses were performed in R using the "phyloseq" package (McMurdie and Holmes, 2013; Callahan et al., 2016b) for data handling and filtering. All Amplicon Sequence Variant (ASVs) assigned to Eukaryota, chloroplast, and mitochondria or not assigned at the phylum level were removed. All ASVs with a total read count lower than 10, across the entire dataset were removed. Also, only features present in at least two samples were retained since two independent DNA extractions were performed on the same biological material, as reported above. All filters were applied before replicate compliance (see below).

Alpha diversity indexes and beta diversities ordination plots were computed with dedicated phyloseq functions. Sample completeness was assessed both using Good's coverage (GC = 1 − F1/N) and iNEXT sample coverage (SC) with the default threshold of SC = 98.5% (Hsieh et al., 2016; Chao et al., 2014). Rarefaction curves, PERMANOVA and ANOVA statistical tests and distance-based redundancy analysis (dbRDA) with capscale function were performed with the "vegan" R package. Differences in alpha diversity metrics were assessed using one-way ANOVA followed by Tukey's HSD *post hoc* test, which includes built-in correction for multiple comparisons (family-wise error rate control). To verify that the data were not affected by compositional constraints, abundances were transformed using centered log-ratio (CLR) and inspected with Aitchison distances. PERMANOVA was used to test differences in community composition (Aitchison distances, 999 permutations), considering fractions, seasonal variation in water samples and spatial variation in sediments. Homogeneity of multivariate dispersion was verified with the betadisper and permutest functions (999 permutations) to ensure that PERMANOVA assumptions were met. dbRDA analysis was performed on bray-curtis distances using the available environment variables described above.

Replicate consistency

As stated above, all samples were processed in replicates starting from the DNA extraction step to sequencing. To assess similarity between replicates a Pearson correlation test was performed on the abundance profile of each identified ASV, p-values were corrected for False Discovery rate. Beta diversity Bray-Curtis dissimilarity plot was also used to visualize the correlation between replicates.

Contamination control

The sediment sampling technique applied did not allow the presence of overlying water to be completely ruled out. To assess potential cross-contamination between water and sediment, a Spearman correlation analysis was performed on the ASV abundance ranks for each water–sediment sample pair. Resulting p-values were adjusted for false discovery rate (FDR) to account for multiple testing.

Functional prediction

Community functional potential was inferred from 16S rRNA gene ASV data using PICRUSt2 v2.5.0 (Douglas et al., 2020), running the standard picrust2_pipeline.py workflow to obtain predicted MetaCyc (Caspi et al., 2014) metabolic pathway profiles for each sample. Given the contrasting ecological patterns observed with metabarcoding – where community differentiation was mainly driven by season in water and by site in sediments – predicted metabolic pathways were analyzed to determine whether they followed the same patterns. Using the ggpicrust2 package (Yang et al., 2023), MetaCyc pathway profiles were summarized and described, and differential abundance analyses of the 15 most abundant predicted pathways were calculated for each month (for water samples) and for each site (for sediment samples).

Identification of discriminant ASVs

Discriminant features were identified using sparse Partial Least Squares Discriminant Analysis (sPLS-DA) from the "mixOmics" R package (v. 6.16.3) (Lê Cao et al., 2016; Rohart et al., 2017). The analysis followed the default MixMC mixOmics pipeline as presented by Calgaro et al., 2021. In brief, ASV count tables were added with a pseudo count of 1, normalized by total sum scaling (TSS), and transformed using the centered log-ratio (CLR). Model performance was evaluated by leave-one-out cross-validation (perf

() function) based on overall and balanced error rates across classification distances, performed separately for water and sediment datasets. Water data were analyzed in search for ASVs that could discriminate and be representative of each season. Sediment samples were analyzed in search for ASVs that could discriminate between the five different stations. For both fractions, the discriminant analysis was applied for each significant interaction, or component, to see the discriminant features that mostly describe each sub-groups (each season or each station). A summary image was plotted for each component displaying the discriminant ASVs loadings, their taxonomy assignments and the related heatmap, using the HotLoadings function of the homonym R package (github.com/mcalgaro93/HotLoadings).

Results

Spatio-temporal trend of environmental variables

Data on water environmental variables can be found in Supplementary Tables 2, 3. No significant differences in temperature were noted between different sites within a given season. However, wide fluctuations in temperature were observed when different seasons were considered. It should be pointed out that Porto Marghera generally recorded the highest temperatures. Salinity, as expected, was higher at the stations located in the inlets (33.3 ± 1.9 ppm and 32.7 ± 1.8 ppm for Lido and Chioggia inlet, respectively). Porto Marghera also showed an average salinity typical of seawater (30.9 ± 2.1 ppm), while Palude della Rosa and San Giuliano recorded a brackish water salinity, and a greater fluctuation in values (28.1 ± 3 ppm and 24.9 ± 4.9 ppm, respectively). Seasonal variations in the percentage of dissolved oxygen (DO) appeared to be dependent on the specific sampling site. Yet, averaging the data by season, the percentage of DO always peaked in winter.

Values of some environmental parameters and micronutrients related to water were only collected during the period from November 2018 to August 2020 (eight seasons). Transparency was generally higher at sites close to the sea, while San Giuliano was nearly always the most turbid, being located in a navigable channel and therefore subject to frequent resuspension of sediment.

Regarding dissolved inorganic nutrients, samples from San Giuliano mainly showed the highest values for nitrogen compounds (DIN value range: 11.27 - 127.46 µM), orthophosphates (0.53 - 2.05 µM) and orthosilicates (30.68 - 91.65 µM) during the study. The lowest values were found in stations located in the inlets (Lido and Chioggia). Furthermore, the lowest concentrations of Chlorophyll-*a* were found in the inlet stations (0.36 – 2.60 µg/L), while some peaks were observed in the other sites during summer (2.99 - 28.07 µg/L).

Values of environmental parameters related to sediments are detailed in Supplementary Table 4. The water depth was the highest at Porto Marghera (11.8 ± 0.6 m) followed by Lido (7.2 ± 0.6 m) while the shallowest occurred at San Giuliano (1.7 ± 0.6 m). Samples from Palude della Rosa showed a higher pH (8.4 ± 0.2) than those from other locations. Regarding surface sediment composition, samples from inlets contained more sand whereas samples from the other locations showed higher percentages of silt and clay.

Sequencing output

DNA samples were sequenced in three Illumina MiSeq 2x300bp runs. The number of retained sequences, after removing primers and too short reads, was 40,008,613 considering the 214 libraries, ranging from 86,570 to 528,691 with a median of 169,305 and an average of 186,956.14 ± 67,886.02. At the end of the dada2 pipeline a total of 21,070,965 non-chimeric denoised sequences were obtained, ranging from 34,418 to 346,933 with an average of 98,462.45 and a median of 86,856 reads per sample (Supplementary Table 5).

Rarefaction curves reached a plateau in all water and sediment samples (Supplementary Figures 1A, B) and coverage-based assessment indicated near-complete sampling for all samples (SC ≥ 0.985), consistent with Good's coverage > 0.99; thus, none of the sample was excluded from further analyses.

Since two replicates were processed for each sample, replicate compliance was assessed with Pearson correlation. R correlation value had a mean value across all replicates of 0.9716 ± 0.0567 (median 0.9806) with a corrected *p*-value lower than 0.001. To visualize the correlation between the replicates, Bray-Curtis beta diversity distances were used to see how close the different replicates were in a dissimilarity plot (Supplementary Figures 2A, B). Since the replicates were confirmed as reliable by Pearson's correlation index and the dissimilarity plot, the abundances of each pair of replicates were summed up, resulting in a total of 107 samples.

Since samples were processed by multiple MiSeq runs, potential batch effects were evaluated. A PCoA based on Bray–Curtis dissimilarities showed no clustering by sequencing run (Supplementary Figure 3). Consistently, PERMANOVA tests on sediment samples and on May water samples (the only water samples sequenced by different runs) showed no significant run effect (sediment: p = 0.133; May water: p = 0.149). These results indicate that sequencing runs did not bias the beta diversity patterns.

Unfortunately, the DNA extracted from the November 2020 Lido sediment sample was not amplifiable, so it was not possible to sequence it. Moreover, as reported in the Material and Method section, the Chioggia sampling station was not sampled at the beginning of the campaign. For these reasons, we do not have a complete time series for all sites.

Microbial community diversity and composition

Alpha diversities

Shannon diversity indexes were significantly different between water and sediment samples being higher for sediment samples (ANOVA test, $F_{1,104}$ = 1689, adjusted p-value < 0.0001, η^2 = 0.94; Tukey's HSD *post hoc* test, adjusted p-value < 0.0001). The same was true also for Evenness index (ANOVA, test $F_{1,104}$ = 577.4, adjusted *p*-value < 0.0001, η^2 = 0.85; Tukey's HSD *post hoc* test, adjusted *p*-value < 0.0001) and Chao1 index (ANOVA test, $F_{1,104}$ = 521.3, adjusted *p*-value < 2e-16, η^2 = 0.83; Tukey's HSD *post hoc* test, adjusted *p*-value 3.508e-14) (Supplementary Figure 4).

Shannon and Evenness alpha diversity indexes revealed spatial differences for water samples (ANOVA test, respectively $F_{4,48}$ = 4.661, adjusted *p*-value < 0.001, η^2 = 0.28 and $F_{4,48}$ = 4.241 adjusted *p*-value < 0.001, η^2 = 0.26). Among the pairwise comparisons, Palude della Rosa had lower values than all other sites according to Shannon index and from Chioggia, Lido and Marghera according to Evenness (Tukey's HSD *post hoc* test, adjusted *p*-value < 0.05 for all comparisons) (Supplementary Figure 5B). Spatial differences were even more marked for sediment samples according to Shannon (ANOVA test, $F_{4,48}$ = 12.41, adjusted *p*-value < 0.0001, η^2 = 0.51) and Evenness indexes (ANOVA test, $F_{4,48}$ = 14.87, adjusted *p*-value < 0.0001, η^2 = 0.55). In the pairwise comparisons, the inlet sites did not reveal any significant differences. Among the two, Lido showed the most marked differences with the inner sites (Tukey's HSD *post hoc* test, adjusted *p*-value < 0.0001 with Marghera and San Giuliano, adjusted *p*-value < 0.05 for Shannon and <0.001 for Evenness with Palude della Rosa). Looking at the inner sites, Palude della Rosa was also considerably different from San Giuliano (Tukey's HSD *post hoc* test, adjusted *p*-value < 0.05 for Shannon) and San Giuliano from Chioggia (Tukey's HSD *post hoc* test, adjusted *p*-value < 0.01 for both Shannon and Evenness) (Supplementary Figure 5D).

In contrast, seasonal differences were marked in the water samples both for Shannon and Evenness indexes (ANOVA test respectively, $F_{3,48}$ = 8.779, *p*-value < 0.0001, η^2 = 0.35, and $F_{3,48}$ = 8.041, *p*-value < 0.0001, η^2 = 0.33), particularly in February samples compared to all other seasons (Tukey's HSD *post hoc* test, adjusted *p*-value < 0.05) (Supplementary Figure 5A). Water samples in August and November showed the highest average values considering the Shannon and Chao1 indexes. This suggests that both species richness and diversity are greater in summer and tend to increase until November. No seasonal variation was observed in the alpha diversity indexes of the sediment samples (Supplementary Figure 5C).

Beta diversity

Prokaryotic communities were significantly different between water and sediment samples (PERMANOVA test, R^2 = 0.28198, *p*-value < 0.0001) (Supplementary Figure 6A). The water microbial communities showed marked seasonal differences (PERMANOVA test, R^2 = 0.44647, *p*-value < 0.0001) (Supplementary Figure 6B), whereas sediment microbial communities showed spatial differences (PERMANOVA test, R^2 = 0.60818, *p*-value < 0.0001) (Supplementary Figure 6C). Homogeneity of multivariate dispersion was verified prior to PERMANOVA: dispersion was homogeneous among months in the water fraction (the main factor analyzed) but differed slightly among sites, whereas in the sediment fraction dispersion was homogeneous for both months and sites. The results indicate that planktonic prokaryotic communities are mainly influenced by seasonal variation rather than geographic distribution, while benthic prokaryotic communities are much more stable throughout the year while changing between sampling sites. The reported R^2 values indicate that a substantial portion of community variation is explained by these factors.

Redundancy analysis

The association between environmental data and the microbial communities was explored by means of redundancy analysis. In particular, a distance-based redundancy analysis was run for both water and sediment samples on the Bray-Curtis distance matrices. The dbRDA analysis on water samples (Figure 2A) revealed that temperature significantly explains most of the differences in community composition (ANOVA *p*-value < 0.001) clearly separating August samples from all the others. Although with lower significance (ANOVA *p*-value < 0.05), reduced nitrogen (NH_3^+) also explains the observed differences, as it seems to influence all November and a few May samples.

The dbRDA analysis on sediment samples (Figure 2B) showed that depth, salinity and sediment composition significantly explain the differences in community structure (ANOVA *p*-value < 0.001), sandy granularity being the major driver of Lido and Chioggia samples, while salinity and depth playing a less pronounced role in driving diversity for some samples of the same sites. In addition, oxygen and silty sediment explain the differences in the sediment samples at a lower significance (ANOVA *p*-value < 0.05), with silty granularity being a driver for all samples from the inner stations, Marghera, San Giuliano and Palude, while oxygen slightly influences the Lido and Chioggia samples.

Taxa abundances

Before applying any abundance filters, we identified 41,166 ASVs. Considering the features that were retained after the abundance filters, we identified 3931 ASVs in the water samples and 19,372 ASVs in the sediments, bringing the total number of unique ASVs to 21,503. Of these, 1147 ASVs (5.3% of the total ASVs identified) were found to be shared between the two matrices when considering water and sediment sampled at the same site and on the same day.

Looking at the whole set of ASVs, 90.1% were found in sediment, whereas 18.28% were identified in water samples (note that the total percentage here is over 100% due to ASVs being found in both matrices). The observed richness in water samples ranged between 243 to 656 ASVs recorded at Lido in May 2021 and San Giuliano in November 2019, respectively (mean 439.7 ± 110.7). In sediment samples, the observed richness ranged from 1467 in Marghera in February 2021 to 4933 in Chioggia in February 2020 (mean 2628.1 ± 706.6) (Supplementary Table 6 for all alpha diversity indexes).

Considering total counts, 195,807 (0.96%) reads were assigned to Archaea and 20,156,740 to Bacteria. As regards the bacterial kingdom, 73 phyla were identified considering all samples, of which 20 were exclusively present in the sediment samples, while the remaining ones were shared between the two matrices.

Figure 3 shows the phyla having a median abundance across all samples, from both sediment and water fractions, higher than 1%. Furthermore, highly variable phyla, those that change the most

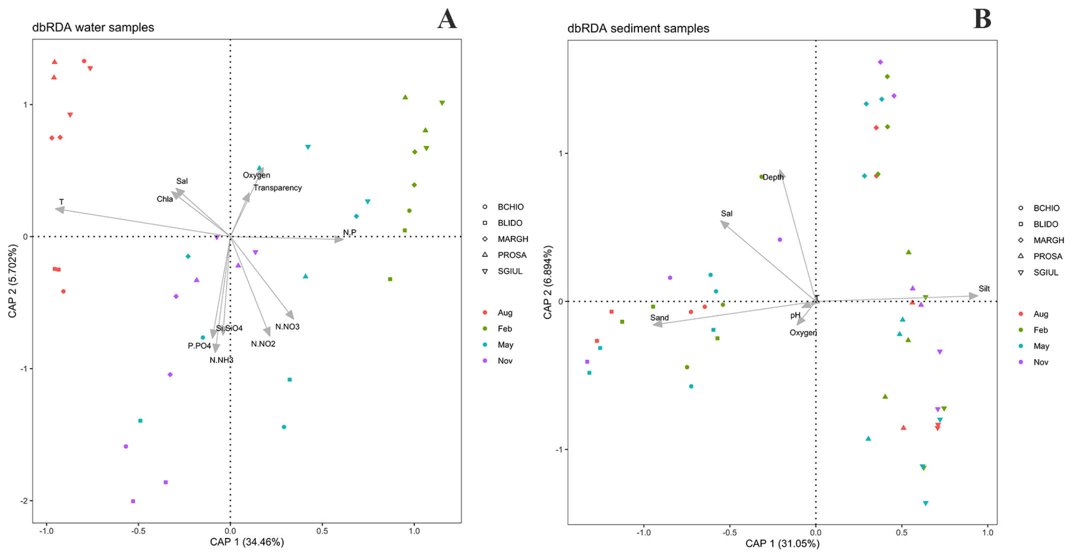

FIGURE 2
(A) dbRDA of environmental variables and water samples. Please note that the graph is related to samples between November 2018 and August 2020, apart from Chioggia site, where water was sampled starting from May 2019. **(B)** dbRDA of environmental variables and sediment samples. Please note that due to the lack of physical parameters for the Chioggia site in November 2018, all samples from that date were excluded.

among samples, are also shown even if their median abundance was less than 1%. All the other phyla (n=63) are grouped in "Other". The number of phyla falling in this category is much higher in sediment samples (63) than in water ones (47). This indicates that many phyla in the sediment domain are rare or present in low abundance.

Proteobacteria dominated all stations and seasons in both matrices, although their abundances were higher in water samples than in sediment samples, averaging 51.97 ± 9.10% and 36.47 ± 3.85% respectively. Besides Proteobacteria, two other phyla were highly abundant in water: Bacteroidota (31.16 ± 7.17%) and to a lesser extent Actinobacteriota (9.21 ± 3.87%). Moreover, Cyanobacteria and Campylobacterota phyla showed high abundances in specific seasons or stations. In particular, Cyanobacteria in Palude della Rosa reached a relative abundance of 21.78% and 17.93% in the two summer samples, an order of magnitude higher than the mean abundance of this phylum in all samples (2.29 ± 4.41%). Similarly, Campylobacterota had an average abundance of 2.41 ± 4.77% but reached 25.24% and 17.42% at San Giuliano in May 2020 and at Marghera in February 2021, respectively.

In sediment samples, in addition to Proteobacteria, which accounted for more than one third of the abundance, other fairly abundant phyla were found: Desulfobacterota (12.30 ± 3.27%), Bacteroidota (15.51 ± 4.68%), and, to a lesser extent, Acidobacteriota (5.22 ± 1.56%), Planctomycetota (4.46 ± 1.89%), Chloroflexi (3.91 ± 1.95%) and Campylobacterota (3.84 ± 2.66%). These last two phyla were generally more represented in the inner sites than in the outer ones (4.61 *vs* 2.8% for Chloroflexi, 5.08 *vs* 1.8% for Campylobaterota), while Actinobacteriota and Planctomycetota showed the opposite trend (3.09 *vs* 0.71% and 6.26 *vs* 3.37%, respectively).

In the water samples, it is interesting to note the high abundance of Cyanobacteria in the inner sites only in August, compared to the other seasons. This is different from what can be seen at the outermost stations where Cyanobacteria are found more or less in all seasons but with a marked increase from May to November, peaking in August, as in the other three stations. These results were also supported by the flow cytometry measurements which showed a similar trend in the number of cyanobacterial cells observed in the same samples (Supplementary Table 3).

Considering both matrices, Figure 4 shows that Actinobacteriota, Bacteriodota, Campylobacterota and Proteobacteria had significantly higher abundances in water than in sediments, while Acidobacteriota, Chloroflexi, Desulfobacterota, Plantomycetota and Verrucomicrobiota had significantly higher abundances in sediments. Cyanobacteria exhibited similar abundances in both fractions, although they were considerably higher in a few water samples, as previously mentioned.

Table 1 shows the most abundant taxa for the lower taxonomic levels of class, order, family, and genus for both water and sediment samples. Regarding water samples, genera with a relative abundance greater than 5% are reported, whereas for sediments, genera above 1% are listed. All the taxa abundances are available in Supplementary Table 7.

Predicted functionals pathways

Based on PICRUSt2 predictions, a total of 434 MetaCyc (Caspi et al., 2014) pathways were identified across all samples. The full list of pathways, together with their abundances in each sample and their descriptions, is provided in Supplementary Table 8.

Across both water and sediment samples, the two most abundant predicted pathways were Aerobic respiration I (cytochrome c) and Gondoate biosynthesis (anaerobic). In water samples (Figure 5A), several MetaCyc pathways were predominantly above the overall

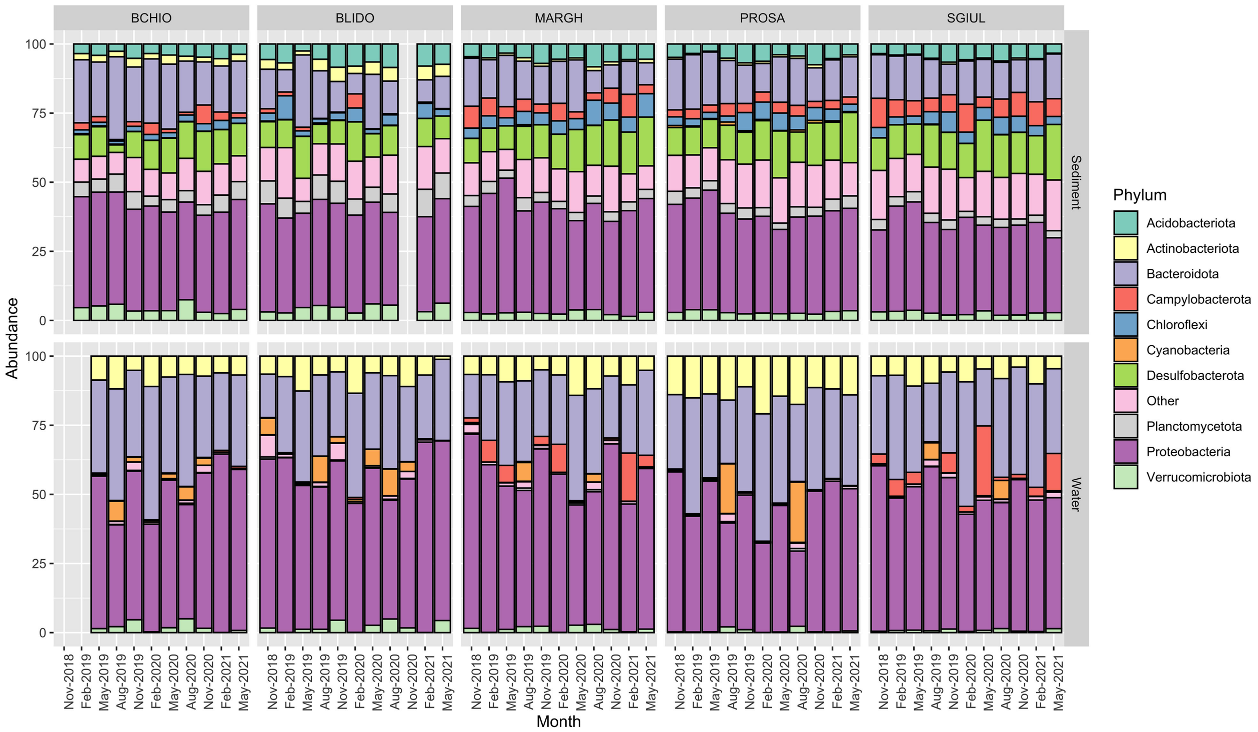

FIGURE 3
Bar chart showing the relative abundance of the most represented phyla in the five different sampling sites and in the two fractions, sediment (upper part) and water (lower part). Sample code names are: Bocca di Chioggia (BCHIO), Bocca di Lido (BLIDO), Marghera (MARGH), Palude della Rosa (PROSA) and San Giuliano (SGIUL).

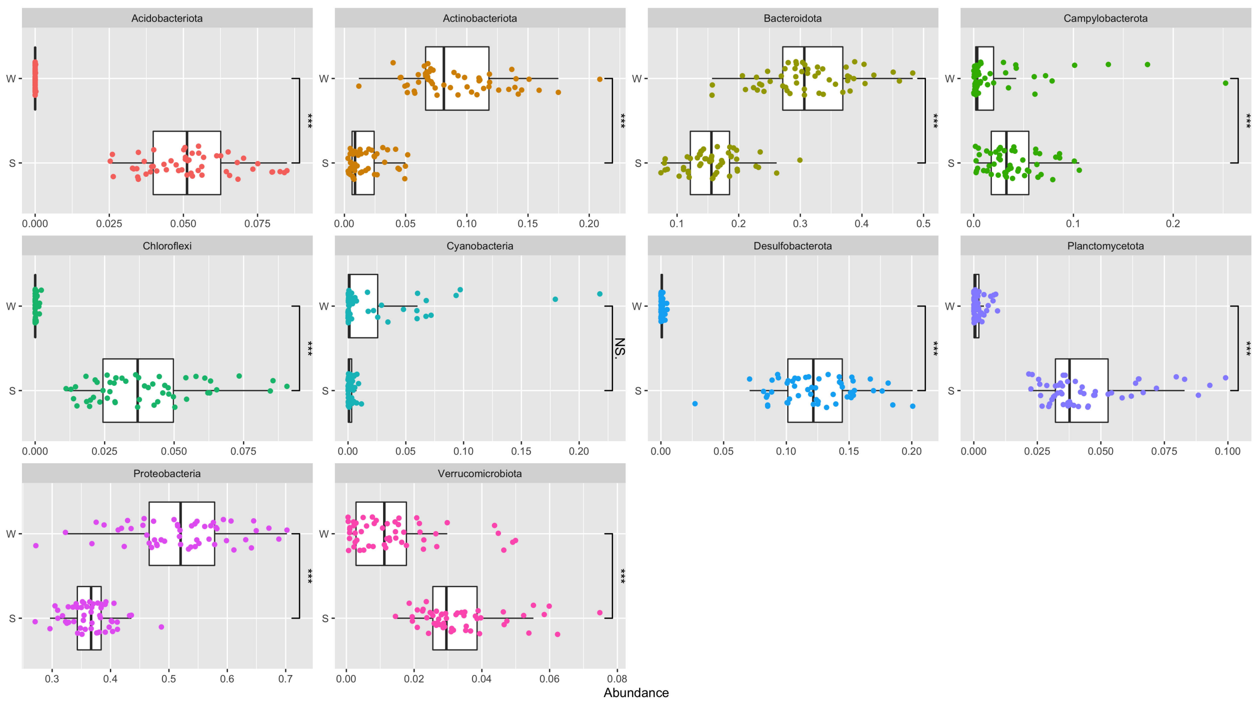

FIGURE 4
Box-plot showing the abundance of the top 10 phyla in all samples. Each point refers to the cumulative abundance of all the ASVs assigned to any given phyla in each sample. Significances according to the ANOVA test between the two groups are shown (*** p-value > 0.001 and N.S. non significant) with W indicating water samples and S indicating sediment ones. Abundances range from 0 to 1.

TABLE 1 Average abundances of the most abundant taxa at taxonomic levels lower than phylum.

<table>
<tr><th>Phylum</th><th>Class</th><th>Order</th><th>Family</th><th>Genus</th></tr>
<tr><td colspan="5">Water</td></tr>
<tr><td>Actinobacteriota
9.21 ± 3.87%</td><td>Actinobacteria
8.19 ± 4.1%</td><td>Micrococcales
7.95 ± 4.1%</td><td>Microbacteriaceae
7.95 ± 4.14%</td><td>Candidatus Aquiluna
6.02 ± 3.54%</td></tr>
<tr><td rowspan="2">Bacteriodota
31.16 ± 7.17%</td><td rowspan="2">Bacteroidia
29.79 ± 7.57%</td><td rowspan="2">Flavobacteriales
26.12 ± 7.64%</td><td rowspan="2">Flavobacteriaceae
19.68 ± 6.85%</td><td>NS3a marine group
5.0 ± 4.67%</td></tr>
<tr><td>NS5 marine group
5.38 ± 2.53%</td></tr>
<tr><td rowspan="3">Proteobacteria
51.97 ± 9.10%</td><td rowspan="2">Alphaproteobacteria
25.45 ± 6.75%</td><td>Rhodobacterales
11.91 ± 6.94%</td><td>Rhodobacteraceae
11.91 ± 6.94%</td><td></td></tr>
<tr><td>SAR11 clade
6.3 ± 4.8%</td><td>Clade I
5.34 ± 4.2%</td><td>Clade Ia
5.32 ± 4.15%</td></tr>
<tr><td>Gammaproteobacteria
26.47 ± 7.32%</td><td>Pseudomonadales
15.15 ± 6.35%</td><td></td><td></td></tr>
<tr><td colspan="5">Sediment</td></tr>
<tr><td>Acidobacteriota
5.22 ± 1.56%</td><td>Thermoanaerobaculia
3.21 ± 1%</td><td>Thermoanaerobaculales
3.21 ± 1%</td><td>Thermoanaerobaculaceae
3.21 ± 1%</td><td>Subgroup 23
2.47 ± 1.17%</td></tr>
<tr><td>Bacteroidota
15.51 ± 4.68%</td><td>Bacteroidia
14.65 ± 4.62%</td><td>Flavobacteriales
5.67 ± 2.49%</td><td>Flavobacteriaceae
5.13 ± 2.22%</td><td></td></tr>
<tr><td>Campylobacterota
3.84 ± 2.66%</td><td>Campylobacteria
3.84 ± 2.65%</td><td>Campylobacterales
3.84 ± 2.65%</td><td>Sulfurovaceae
2.96 ± 2.25%</td><td>Sulfurovum
2.96 ± 2.25%</td></tr>
<tr><td>Desulfobacterota
12.30 ± 3.27%</td><td>Desulfobacteria
6.50 ± 2.93%</td><td>Desulfobacterales
5.59 ± 2.57%</td><td>Desulfosarcinaceae
4.71 ± 2.06%</td><td>Sva0081
2.13 ± 0.85%</td></tr>
<tr><td rowspan="2">Proteobacteria
36.47 ± 3.85%</td><td rowspan="2">Gammaproteobacteria
33.36 ± 3.29%</td><td>Steroidobacterales
6.02 ± 1.44%</td><td>Woeseiaceae
5.96 ± 1.47%</td><td>Woeseia
5.86 ± 1.34%</td></tr>
<tr><td>Pseudomonadales
7.28 ± 2.17%</td><td>Halieaceae
5.83 ± 2.11%</td><td>Halioglobus
3.79 ± 1.56%</td></tr>
</table>

mean in August and November, including Aerobic respiration I, L-isoleucine biosynthesis and L-valine biosynthesis. These pathways represent core heterotrophic functions linked to oxidative energy production and anabolic amino acid synthesis. Overall, pathway distributions showed a clear seasonal pattern, with lower relative abundances in February and May and higher values in August and November. November displayed the highest number of pathways above the mean, whereas February and May showed the opposite trend, mirroring DNA metabarcoding results, where communities were depleted in February, partially recovered in May, peaked in August and remained diverse and active in November.

In sediment samples (Figure 5B), predicted pathway abundances reflected a clear spatial structure. The inlet sites showed similar functional profiles and contained most pathways above the mean, whereas the inner sites generally deviated less from the mean. The incomplete reductive TCA cycle and L-isoleucine biosynthesis V were notably more abundant in Palude della Rosa, Marghera and San Giuliano, reinforcing the functional distinction between inner and outer lagoon sites, consistent with the site-specific clustering observed via DNA metabarcoding. Among the most abundant pathways, cis-vaccenate biosynthesis – producing the monounsaturated fatty acid cis-vaccenate, a common bacterial membrane component – and the prokaryotic TCA cycle, a core central carbon and energy pathway, stood out, with the TCA cycle showing the smallest variation in relative abundance across sites.

ASVs shared between water and sediment

To assess potential overlap between water and sediment microbial communities due to cross-contamination during sediment sampling, all sample pairs were tested using Spearman's correlation. After false discovery rate (FDR) correction, no statistically significant correlation was found (Supplementary Table 9). This rules out contamination between water and sediment samples as a major driver in shaping the sediment prokaryotic communities. Looking at each couple of water and sediment samples collected in the same day, 1147 ASVs were identified as shared between the two matrices in at least one couple. Shared ASVs represent 5.3% of total identified ASVs but it is worth noting that they account for almost a third of aquatic ASVs (29.1%) while less than 6% of sediment ASVs. More interestingly, these shared ASVs were found to collectively represent 72.61 ± 6.97% of total abundances in water samples and 31.77 ± 7.81% in sediments. A small number of these ASVs, 37, were shared by only a single pair of water-sediment samples, while

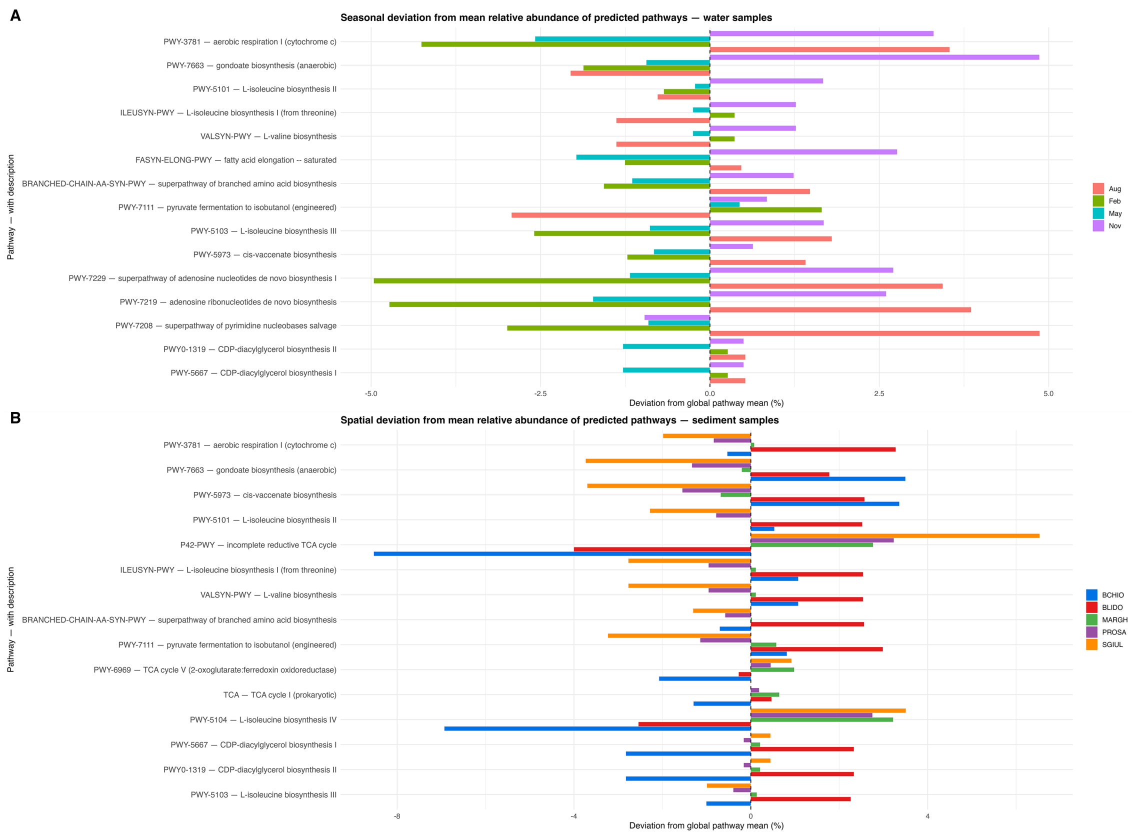

FIGURE 5
Seasonal **(A)** and site **(B)** variation in the relative abundance of predicted pathways. Values show the percentage deviation of each pathway from its global mean relative abundance. Bars represent the 15 most abundant pathways across months **(A)** or sites **(B)**, while the y-axis lists the pathway names together with a short functional description. The pathways are listed in order of abundance in the dataset, from the most abundant to the 15 most abundant.

75% were shared between at least 8 pairs. No significant differences were found at any site in terms of the number of ASVs shared between water and sediment.

Most of the ASVs shared between the two matrices belong to Proteobacteria, Bacteroidota and Verrucomicrobiota (Supplementary Figure 7). Some other less represented phyla are Plantomycetota, Campylobacterota and Bdellovibrionota (Supplementary Table 10). The average abundance of these ASVs was low, being 0.04 ± 0.17%. However, some ASVs were much more abundant, with nine ASVs having an average abundance of more than 1%. A comparison of the abundances of these nine features in the two matrices revealed that they were significantly more abundant in water samples than in sediments (data not shown).

Among the whole set of shared ASVs, fifty have been assigned at the species level. Supplementary Figure 8 shows the heatmap of the abundances of these features. Interestingly, although all of these were found to be shared in at least one couple of water and sediment samples, many appear to be peculiar to either the sediment or water fraction. For example, ASV87, assigned to *Eudoraea adriatica*, appears to be mainly present in sediments. The same seems to be true also for ASV238 (*Lutimonas saemankumensis*), ASV709 (*Sulfitobacter brevis*), and ASV433 (*Roseibacterium elongatum*). Some others, on the contrary, appeared to be mainly present in water samples, such as ASV2 (*Planktomarina temperata*) and ASV118 (*Amylibacter ulvae*). Finally, some were confirmed to be present in both fractions in a similar fashion, as for example ASV2168 (*Psychrilyobacter atlanticus*), ASV779 (*Fabibacter halotolerans*), ASV1538 (*Hellea balneolensis*), ASV75 (*Helicobacter pylori*), and ASV166 (*Arcobacter cryaerophilus*).

Archaeal community

Looking at all samples, 1430 ASVs were assigned to Archaea. In the whole dataset, the cumulative abundance of these features was 0.96%, while the mean abundance per sample was 0.84 ± 0.9%, being 0.22 ± 0.48% in water and 1.46 ± 0.82% in sediments. These taxa belong to ten phyla, which were differently represented in the two fractions (Supplementary Figure 9). The most abundant phyla in the sediments were Crenarchaeota, Nanoarchaeota and Thermoplasmatota, and, to a lesser extent, Asgardarchaeota and Halobacterota. In particular, Crenarchaeota was the most abundant archaeal phylum at Lido and Chioggia stations while at Marghera, San Giuliano and Palude della Rosa Crenarchaeota, Nanoarchaeota and Thermoplasmatota were more similar in abundance. Sediments from the Lido site in February 2019 showed the highest abundance of Archaea, reaching 5% of the total counts for that particular sample.

In water samples, almost all archaeal taxa were found to be in low abundance. In fact, only the phylum Thermoplasmatota was found to reach almost 2.5% of the total counts in November 2019 at the Lido site. Archaeal ASVs were almost absent in Palude della Rosa and San Giuliano water samples.

Bacterial indicator community

Data were analyzed looking for potentially harmful bacteria such as those belonging to Arcobacteraceae family (Venâncio et al., 2022) and those of potential human and non-human fecal sources such as Enterobacteriaceae, Bacteroidaceae, Clostridiaceae, Lachnospiraceae, Porphyromonadaceae, Prevotellaceae, Rikenellaceae and Ruminococcaceae families (Newton et al., 2013, 2015; Basili et al., 2022). These families showed abundances below 1% in all samples except for Bacteroidaceae and Arcobacteraceae. In fact, the San Giuliano's water sample of May 2021 showed abundances of 1.7% for Bacteroidaceae. This result was due to several ASVs, the most abundant of which, at 0.89%, was assigned to *Bacteroides graminisolvens*, a bacterium first isolated in bioreactors for processing cattle waste, where it is involved in breaking down complex carbon compounds (Nishiyama et al., 2009).

Among the entire dataset, 86 ASVs were assigned to the Arcobacteraceae, which emerged as the main bacterial indicator family in terms of ASVs and abundances. A more in-depth analysis revealed that the cumulative abundance of this family was less than 0.5% in the sediments. Similarly, Arcobacteraceae had an abundance of less than 1% in 75% of the water samples. Interestingly, all the samples in which this family was reported at abundances greater than 1% were from water fractions of the San Giuliano and Marghera sites. In particular, bacteria belonging to this family accounted for 15.6% of the total counts at Marghera in February 2021 and for 11.4% at San Giuliano in May 2021.

Among the 86 ASVs, 54 were not assigned at the genus level, while the others were assigned to *Halarcobacter* (14), *Pseudarcobacter* (9), *Arcobacter* (4), *Malaciobacter* (3) and *Poseidonibacter* (2). Some of these were identified up to the species level: *Arcobacter cryaerophilus*, *A. cibarius*, *A. butzleri*, *Pseudoarcobacter suis*, *P. aquimarinus*, *P. ellisii*, *Poseidonibacter lekithochrous*, *Halarcobacter bivalviorum* and *Malaciobacter pacificus*.

With regard to three species identified within the *Arcobacter* genus, it must be clarified that their taxonomic assignments depend on the SILVA database version, as recently these three species were moved to the new genus *Aliarcobacter* (Pérez-Cataluña et al., 2018, 2019). Moreover, a BLAST search (Zhang et al., 2000) against the Prokaryota experimental taxonomic nt database at NCBI of the fourth feature assigned to the *Arcobacter* genus, ASV121, identified it as an *Aliarcobacter* species with 100% identity (data not shown) with both *Aliarcobacter faecis* strain CCUG 66484 and *Aliarcobacter lanthieri* strain LMG 28516. As previously mentioned, the Marghera water sample from February 2021 had the highest total abundance of Arcobacteraceae features identified among all samples, accounting for 15.66% of the total counts. Six of the 86 features had abundances greater than 1%, totaling 11.5% of the sample's counts: ASV928, ASV229 and ASV664 were assigned only at family level to Arcobacteraceae, while ASV1038 was assigned to the *Pseudarcobacter* genus. Finally, the features with the highest abundances were ASV185 (2.88%) and ASV230 (4.1%), the former being annotated as *Pseudarcobacter suis*, while the latter as *Pseudarcobacter* sp.

In the San Giuliano sample of May 2021 there were only four features above 1% abundance: ASV432 (*P. aquimarinus*), ASV185 (*P. suis*), ASV166 (*Arcobacter cryaerophilus*) and ASV121, which had the highest abundance (3%) and, as reported above, was assigned to *Aliarcobacter faecis*.

Discriminant features

Based on the results of the beta diversity analysis (see above and Supplementary Figure 6), discriminant features were searched for seasonal variation in water and for spatial variation in sediment samples. To do so the sparse Partial Least Squares Discriminant Analysis (sPLS-DA) was performed to identify the discriminant ASVs, i.e. those that most characterize each season in the water fraction or each site in the sediment matrix. Model validation by leave-one-out cross-validation confirmed the robustness of the discriminant analyses. The overall classification error rate was 0.17 for the water dataset and 0.04 for the sediment dataset (centroid distance), indicating a stable and reliable discrimination among groups.

Discriminant features in water samples

Four main components in the water fraction were able to discriminate between the four seasonal clusters. The first component mainly contributes to the differences between August and February samples, so most of the identified discriminant features are characteristics of these two seasons. However, some features of this component are also peculiar to May and November samples. We found 47 features characterizing August samples, 18 features characterizing February samples, 15 features characterizing November samples and 5 characterizing May samples. The features that discriminate between August and November samples appear to be exclusive to those two seasons as they show really low abundance in February and May samples. Among them, it is worth mentioning the presence of ASVs belonging to the genera *Synechococcus* in August samples and *Cyanobium* in November samples.

The remaining components mainly help to discriminate between the mid-seasons even though the fourth component identified one additional feature that characterizes August samples and two that characterize November samples. These features are not strictly present only in these seasons, as in the case of the features identified with the first component, but are also present in other seasons, albeit to a lesser extent. Thus, considering all the components analyzed, the total number of features characterizing the different seasons was 48 for August, 24 for November, 20 for February and 8 for May (Supplementary Table

11). The most represented phyla among the discriminant features found in water were Proteobacteria (53%) and Bacteroidota (35%), which collectively account for 88% of the ASVs identified with this analysis. Looking at the family level, the most represented were Flavobacteriaceae, SAR116 clade and Rhodobacteraceae.

Discriminant features in sediment samples

In sediments, the multivariate analysis identified four main components that were able to effectively discriminate among the five stations. The first component discriminates between sites close to the sea and those located further inland in the lagoon. All the 20 features identified in analyzing this component are peculiar to both Chioggia and Lido stations; among these, three seem to be mostly associated with Chioggia samples.

The second component mainly discriminates toward Marghera and Chioggia stations with two ASVs that are more peculiar to Chioggia and two to Marghera. These four ASVs however are present in both sites, although with different footprints. A fifth ASV in this component is identified to be peculiar to Lido sediments but the same feature is also present in Palude and San Giuliano. The third component mainly identifies features associated with sediments of Palude della Rosa. Among the 20 discriminant features, 18 are peculiar to Palude sediments while one is characteristic of San Giuliano, even though it is also present to a lesser extent in other stations (Marghera, Lido and Chioggia where it is particularly rare). One feature mainly characterizes Chioggia, although it is also present in Palude. The fourth component identified discriminant features between San Giuliano, 4 ASVs, and Lido, 1 ASVs (Supplementary Table 11 for a list of discriminant features). The most represented phyla in all the discriminant features found in sediments were Proteobacteria (34%), Bacteroidota (28%) and Acidobacteriota (10%). Among the ASVs assigned to the family level, Woesiaceae (10%) and Flavobacteriaceae (8%) were the most abundant.

Discussion

Coastal lagoons are ecotonal systems that provide key ecological services, including nutrient cycling, decomposition, shoreline stabilization, and the regulation of fluxes of nutrients, sediments, and organisms between land and sea (Levin et al., 2001; Danovaro and Pusceddu, 2007). Their hydromorphology leads to strong spatial and temporal variability of environmental conditions (Pérez-Ruzafa et al., 2011). At the same time, lagoons are subject to intense human pressures due to industrial, commercial, and recreational activities (Solidoro et al., 2010). A well-known example is the Venice Lagoon (Italy), the largest coastal transitional ecosystem in the Mediterranean, heavily shaped by human interventions over the centuries (Madricardo and Donnici, 2014). The most recent of these is the MOSE system, a set of mobile barriers designed to temporarily isolate the lagoon from the Adriatic Sea during high tide events, to prevent flooding (Trincardi et al., 2016). This system constitutes a substantial hydrodynamic perturbation for the lagoon, with potential repercussions on biogeochemical processes, sedimentary regimes, and microbial community dynamics (Del Bello, 2018; Viero and Defina, 2016; Tognin et al., 2020). Assessing the ecological consequences of these episodic closures requires robust baselines. This study provides a seasonal, multi-annual characterization of water and sediment prokaryotic communities, offering critical reference data for future evaluations of MOSE-induced shifts in microbial ecosystem structure and function.

Prokaryotic community diversity, composition and function

Overall, this research shows that water and sediment samples host very different microbial communities (Figure 3; Supplementary Figure 6A). Furthermore, as reported in previous studies (Ul-Hasan et al., 2019; El-Malah et al., 2023), the marine prokaryotic community showed greater diversity in sediments than in seawater samples (Supplementary Figure 4). The alpha and beta diversity indices also suggest that seasonal variations have little impact on sediment communities (Supplementary Figure 5C) (Miksch et al., 2021), which are clearly isolated from one another (Supplementary Figure 6C). Resuspension events therefore do not appear to transport sediment-associated microbes very far from their origin. On the other hand, the data suggests that water microbial communities are more homogeneous throughout the lagoon and are mainly influenced by seasonal variation rather than sampling location (Supplementary Figure 6B) (Celussi et al., 2009; Ferrera et al., 2024). It is likely that the high rate of water mixing, mainly due to the daily frequency of tidal influences (Cucco and Umgiesser, 2006), contributes to a homogeneous microbial community throughout the lagoon system, or at least at the sampling stations considered.

Functional inferences, predicted using PICRUSt2 (Douglas et al., 2020) and MetaCyc (Caspi et al., 2014) were used to test whether the metabolic potential of the microbial community mirrored the seasonal (water) and spatial (sediment) patterns observed with DNA metabarcoding. Across both compartments, Aerobic respiration I (cytochrome c) and Gondoate biosynthesis (anaerobic) were the most abundant pathways. Other abundant pathways included cis-vaccenate and diacylglycerol biosynthesis, as well as several amino acid biosynthesis pathways. These functions represent core metabolic functions of microbial communities, providing energy through aerobic respiration and supplying membrane lipids and amino acids required for growth. In the water, pathway abundances varied seasonally, in line with taxonomic patterns. Growth- and energy-related functions, such as Aerobic respiration I and branched-chain amino acid biosynthesis (L-isoleucine and L-valine), peaked in August and remained high in November, whereas most pathways were reduced in February and only partially recovered in May. One exception was pyruvate fermentation to isobutanol, which was exclusively predicted in the water samples and showed a marked decrease in August. This pattern may reflect the fact that in August the planktonic community is more strongly dominated by phototrophs and aerobic growth-related pathways, making this specific fermentative route relatively under-represented compared with the other months. All these findings support the view of a seasonally responsive and metabolically flexible planktonic community. In sediments, dominant predicted functions included membrane lipid biosynthesis, amino acid biosynthesis and carbon fixation. As observed for the taxonomic composition, functional profiles also separated the inner lagoon sites (Palude della Rosa, Marghera and San Giuliano) from the more seaward sites (Lido and Chioggia inlets). The incomplete reductive TCA cycle, predicted only in sediment samples, was less represented at the inlet sites and more abundant at the inner sites. Although incomplete, this pathway can convert pyruvate into key biosynthetic intermediates under anaerobic or microaerophilic conditions. Its distribution mirrors the sediment structure, with inner sites consisting of a clayey silt layer that provides a favourable niche for anaerobic carbon-fixation processes.

Water-sediment shared ASVs

The deployment of Van Veen grabs for sediment sampling is subject to a certain degree of contamination from the overlying water. Consequently, an analysis was performed to ascertain whether this contamination could affect the structure of the microbial communities found in the sediments. However, this was found not to be the case. The 1147 ASVs found in both water and sediment samples, and the evidence that they collectively represent a good fraction of the total microbial abundance in water and sediment, raises the question of whether or not these prokaryotes are adapted to both environments. A possible explanation for the degree of shared ASVs could be the shallow depth of the lagoon and the resuspension of sediments due to water movement caused by ship and boat traffic, tides, river inflows and other natural currents (Rapaglia et al., 2011). If this were the case, one would expect shallow sites, such as San Giuliano and Palude della Rosa, to have a much higher percentage of shared ASVs than the deeper ones, Marghera and Lido. The fact that there is no correlation between the sites and their depth, and the amount of ASVs shared between surface water and sediment, rejects this hypothesis, indicating that depth has no influence on the presence of shared ASVs. Also, these shared features have a higher average abundance in the water fraction than in the sediment, thus this could not be explained by the resuspension of a portion of the sediment from the lagoon bottom.

However, the fact that many of these features were significantly more abundant in water samples than in sediments raises another hypothesis: could this indicate that these ASVs are only present in sediments because they sink from above? Even if this were the case, it does not fully explain what was observed. Sediment and water have a very different number of identified ASVs, so the abundance of a single ASV found in both matrices could not be of the same extent and therefore cannot be easily compared. In the present study, only one third of the aquatic features were reported in sediments. Future studies could investigate whether or not these shared ASVs are vital and active in both matrices by designing proper biochemical and metatranscriptomic analysis (Davey and Guyot, 2020; Vuillemin et al., 2022).

Among the set of shared ASVs, some were assigned at the species level (see Results). ASV2168 was assigned to *Psychrilyobacter atlanticus*, which was first isolated from marine sediments (Zhao et al., 2009) and was reported to be an obligately anaerobic marine bacterium. Species of the genus *Psychrilyobacter* are preferentially associated with the gut of some marine invertebrates, such as abalone (Liu et al., 2023) and the deep-sea snail *Rubyspira osteovora* (Aronson et al., 2017). At the same time, free-living members of this genus have been reported in both anoxic marine waters and marine sediments, albeit with extremely low abundances (Yadav et al., 2021; Zhao et al., 2009), so it was surprising to see this ASV in the lagoon oxic water samples. Although no conclusions can be drawn about the respiration metabolism of ASV2168 with the data presented in this work, it is also possible that this feature is a close relative of *P. atlanticus* that is not strictly anaerobic. This hypothesis is supported by the finding of aerotolerance-related genes in the genomes of the closely related free-living *P. piezotolerants* strains $SD5^T$ and BL5 (Yadav et al., 2021).

Fabibacter halotolerans (ASV779) was firstly identified from a marine sponge (Lau et al., 2006) and reported to be strictly aerobic and chemo-organotrophic. This feature, as well as related species, is known to establish mutualistic interactions with both diatoms and dinoflagellates (Di Costanzo et al., 2023; Takagi et al., 2023). The fact that this ASV is most abundant in spring and summer water samples seems to correlate with the abundance pattern of these phytoplanktonic taxa in the Venice Lagoon (Bernardi Aubry et al., 2021).

Hellea balneolensis (ASV1538) was first isolated from surface waters of the north-western Mediterranean Sea (Alain et al., 2008) and reported to be an aerobic, heterotrophic, prosthecate bacterium. The *Hellea* genus has been identified as a member of the epiphytic bacterial community of *Ulva* spp (Juhmani et al., 2020; Comba González et al., 2021). Interestingly, this ASV was found in all winter water samples, a fact that correlates nicely with the seasonal increase of *Ulva* in Venice lagoon (Solidoro et al., 2010).

ASV166 (*Arcobacter cryaerophilus*) will be discussed later, as it belongs to the fraction of taxa identified as sewage-associated microbial indicators in these environments.

Among the features shared by water and sediment samples, ASV146 was the one shared by the largest number of samples, 92 out of 106, with an average abundance of 0.09 ± 0.14%. This ASV was assigned to the archaeal genus *Candidatus Nitrosopumilus*, confirming the ubiquity of the ammonia-oxidizing archaea in the environment due to their pivotal role in nitrification (Alves et al., 2018). This ASV was most abundant in the samples from the sites near the sea (Lido and Chioggia) and in two samples from Marghera.

Archeal community

The archaeal community was not addressed in the previous study that investigated planktonic and benthic bacterial diversity of the Venice lagoon over a one-year period (Quero et al., 2017).

In that survey, archaeal sequences were reported with a lower average abundance than in the present dataset (on average <0.2% per sample versus 0.8% per sample, respectively). This difference can mainly be attributed to the primer pair used in the previous work (341F/785R), which gives a high overall coverage of the bacterial domain, but is not equally efficient for the archaeal one (Klindworth et al., 2013). Similarly, the mean abundance of ASVs assigned to Archaea in the sediment samples of the present study was slightly lower than that reported by Banchi et al., 2021 (1.46% versus 3.8%, respectively). This discrepancy can again be explained by the use of a different primer pair (515F-Y/926R in the study by Banchi and colleagues). Indeed, the amplification of a staggered mock community shows that the observed percentage of archaeal taxa obtained with the 515F-Y/806R pair is lower than that obtained with the 515F-Y/926R pair, which is sometimes higher than expected (Parada et al., 2016).

The high abundance of the phylum Crenarchaeota in sediments was somehow expected according to the results from Banchi et al. (2021) and the known ubiquitous presence of members of the Miscellaneous Crenarchaeota Group in marine sediments (Kubo et al., 2012) (Supplementary Figure 9). However, the relatively high abundance of Nanoarchaeota in the innermost sampling sites of the Venice lagoon had not been reported previously.

Among the Crenarchaeota, the most common taxon was *Candidatus Nitrosopumilus* (Walker et al., 2010). This genus was found to be particularly abundant in the more oxic, sandy sediments of Lido and Chioggia, where it serves as an aerobic ammonia-oxidizing chemolithoautotrophic archaeon (AOA). The analysis of the relative abundance of ammonia-oxidising archaea and bacteria suggests that *Candidatus Nitrosopumilus* is responsible for most of the ammonium oxidation in the assessed sediment and water samples (data not shown).

In the anoxic sediments of Marghera, San Giuliano and Palude della Rosa, the Crenarchaeota were represented by the Bathyarchaeia, which formed metabolically interconnected communities with other archaeal groups identified in these sites. The Bathyarchaeia and the Lokiarchaeota, the latter belonging to the Asgardarchaeota phylum, degrade organic compounds to produce H_2 and/or acetate, and fix CO_2 via the Wood–Ljungdahl pathway (Nesbø et al., 2024; MacLeod et al., 2019). This pathway is also used by the Marine Benthic Group D of Thermoplasmatota, which additionally remineralizes proteins and assimilates sulphur, enabling them to exploit diverse benthic carbon sources (Zhou et al., 2019). While lacking a complete TCA cycle and possessing limited biosynthetic capacity, the Woesearchaeales have the potential for acetate fermentation and hydrogen metabolism. This group of Nanoarchaeota establishes a syntrophic partnership with the Halobacterota, anaerobic methanogenic archaea that utilize H_2/CO_2 or acetate (Liu et al., 2018), obtaining essential metabolites to compensate for their metabolic deficiencies. Another group of Halobacterota was the anaerobic methanotrophic archaea (ANME), which cooperate with sulphate-reducing bacteria (SRB) in sulphate-dependent anaerobic oxidation of methane (S-AOM) (Wallenius et al., 2025).

In line with previous studies, Marine Group II (MGII), within the Thermoplasmatota phylum, was found to be the most abundant group of planktonic archaea in water samples (Rinke et al., 2019). These organisms exhibit a photoheterotrophic lifestyle, combining phototrophy via proteorhodopsins with the remineralization of high-molecular-weight organic matter (Tully, 2019), therefore play an important role in the global carbon cycle.

Bacterial indicator community

Among the aquatic microbial organisms, some can indicate contamination by wastewater from human activities and can be dangerous for health. Although this study did not aim to assess the extent of these contaminants at the sampling sites, their presence was nevertheless investigated. To this end, the abundance of ASVs belonging to potentially harmful bacteria was investigated. In fact, some species of the Arcobacteraceae family, which are frequently found in aquatic environments, are known to be associated with human or animal diseases (Venâncio et al., 2022). In addition, the presence of the Enterobacteriaceae, Bacteroidaceae, Clostridiaceae, Lachnospiraceae, Porphyromonadaceae, Prevotellaceae, Rikenellaceae and Ruminococcaceae families, which have been commonly reported in human and non-human fecal sources, was as well investigated (Newton et al., 2013, 2015; Basili et al., 2022).

Having two and a half years of sampling at five different sites in the Venice lagoon, covering the four sub-basins into which it is hydrodynamically divided, we aimed to measure the occurrence of microbial pollutants in this environment. Among the potential signatures of microbial pollution, sewage-associated bacterial taxa were found to be the most abundant in the lagoon, with the genus *Arcobacter* reaching an overall average relative abundance of 0.5%. A recent work has focused specifically on microbial pollution in the Venice lagoon (Basili et al., 2022). The two datasets are actually difficult to compare, as Basili and colleagues 1) sampled on a single day from several sites in the central sub-basin alone (from Porto Marghera to the open sea via Venice), 2) divided each sample into particle-attached and free-living fractions, and 3) amplified the DNA with a different primer pair. The study also shows that the presence and distribution of bacterial indicator taxa changes depending on the sampling location, which further complicates any comparison of data. These are probably the reasons why the relative abundance of traditional and alternative feces-associated taxa is very low in our data, and why *Arcobacter* predominates over the other two sewage-related taxa, *Acinetobacter* and *Trichococcus*, in contrast to the findings of Basili and colleagues (Supplementary Table 7).

In our data the two sampling sites with the highest abundances of the *Arcobacte*r genus, Marghera and San Giuliano, are those closest to human activities. Marghera is in the middle of Venice's industrial port, while San Giuliano is on a canal that runs out of Mestre, Venice's mainland suburb. The presence of the Arcobacteraceae family, which includes some species associated with human and animal diseases, may indeed be favored by industrial pollution loads, poor water treatment facilities and rapid urbanization rates (Venâncio et al., 2022).

Discriminant features for water and sediment samples

The analysis with the MixMC mixOmics pipeline was performed to retrieve information about the most discriminant features (i.e. ASVs) that in the present study are key microbial organisms associated with a given habitat or environment. We identified the organisms that were most characteristic of the different seasons, in the water, and of the different sites, in the sediments (Supplementary Table 11).

Seasonal specific features in water

Some features characterizing water seasonality were found to be more interesting. ASV62 was assigned to the genus *Polaribacter* and was found to be a discriminant feature of February samples. This genus was firstly described in Arctic and Antarctic sea ice (Gosink et al., 1998), but many other species of this genus have been reported at lower latitude and in more temperate waters between 15 and 17°C (Nedashkovskaya et al., 2005; Fukui et al., 2013). Also, *Polaribacter lacunae*, a member of this genus, was found in a lagoon ecosystem in Korea during winter (Kang et al., 2017). Furthermore, this genus was reported to be part of the seaweed associated microbial community in the Venice lagoon (Juhmani et al., 2020) and found to be present in the northern Adriatic Sea in March (Trano et al., 2022). As reported in literature, this genus is adapted to cold water. This is also confirmed by the result presented in this study, as ASV62 was reported to be highly abundant in February samples, when the average temperature is 9.69°C. Interestingly, even though the average temperature in November (16.44°C) is lower than in May (18.85°C) (Supplementary Table 2), this ASV was found to be present in some May samples and almost absent in November ones. It has recently been suggested that during spring algal blooms in the southern North Sea, most of the remineralization of algal glycans is mediated by only a dozen dominant Bacteroidetes clades, including prominently *Polaribacter* (Krüger et al., 2019). It is therefore possible that the occurrence of this genus observed in the present study is related to its function as an initial degrader of high molecular weight algal polysaccharides (Teeling et al., 2012; Buchan et al., 2014).

One of the features identified as discriminant for May was ASV102 that was assigned to *Nereida ignava*. This species was first identified in the waters surrounding cultivated oysters off the coast near the city of Valencia in Spain (Pujalte et al., 2005). In our study, this ASV was identified in all May samples and to a lesser extent in February samples. Genome analysis of this bacterium revealed the presence of genes and pathways with potential activity in the cycling of carbon, nitrogen and sulphur, as well as in the utilization of algae-derived compounds (Arahal et al., 2016). This suggests that ASV102 is part of the swift successions of distinct bacterial clades that are driven by the spring phytoplankton blooms (Teeling et al., 2016).

ASV189 was found as a discriminant feature of August, but was also present, albeit to a lesser extent, in November and May. Despite its low abundance in all sites (average of 0.09%), this ASV is of high relevance because it was assigned to the *Alcanivorax* genus, which includes species able to degrade alkane compounds, hydrocarbons, oil and plastics as for example *Alcanivorax borkumensis* (Yakimov et al., 1998; Davidov et al., 2020; Koike et al., 2023). ASV189 appears to be widely distributed throughout the lagoon, with no specific location where it might be more abundant. As it also occurs in other seasons, it can be assumed that this organism is always present in the lagoon, although it becomes slightly more abundant in summer.

ASV375 was assigned to the *Cyanobium gracile* PCC 6307, a strain belonging to the genus *Cyanobium*, which comprises non-marine photoautotrophic picoplankton within the *Cyanobacteria* phylum (Callieri et al., 2013). This ASV emerged as a discriminant feature in November, when it was present at high levels in most samples, with the exception of Palude della Rosa and San Giuliano in 2020. Moreover, the same ASV had high abundances in August samples of Lido, Chioggia and Palude della Rosa.

Spatial specific features in sediments

A variety of factors explains some of the discriminant features found in the sediment samples. Firstly, five different features belonging to the genus *Woeseia* were identified in Chioggia (ASV532 and ASV425), Lido (ASV372) and Palude della Rosa (ASV3052 and ASV581). Although *Woeseia oceani* is the only species of this genus that has been isolated and fully described to date (Du et al., 2016), it is now known that the Woeseiaceae bacterial family represents a group of Gammaproteobacteria that are globally distributed and abundant in both coastal and deep-sea sediments (Hoffmann et al., 2020). The functional analysis of Woeseiaceae MAGs has revealed that these widespread taxa are highly metabolically flexible (Chen et al., 2021). Furthermore, coastal and especially deep-water sediments usually harbor a mix of Woeseiaceae taxa, whose co-existence suggests that they have different ecological functions (Hoffmann et al., 2020). This seems to explain the detection of two discriminant features assigned to genus *Woeseia*, both in Chioggia and Palude della Rosa. The ASVs of Palude della Rosa also confirm the widespread presence of the *Woeseia* genus. ASV581 was found distributed across most sites, with higher abundance in Palude della Rosa sediments, but absent from San Giuliano. Conversely, ASV3052 was found to be very abundant in Palude della Rosa and much less abundant, but always present, in all the other sampled sites.

Some ASVs belonging to the genus *Woeseia* were also reported in the sediments of Marghera, although none were found to be a discriminant feature of this sampling site. This finding was somewhat unexpected, as the industrial port area, where this site is located, has been described to be highly contaminated with hydrocarbon derivatives (Cassin et al., 2018) and it has been previously reported that a metagenomic bin associated with *Woeseia* harbors the metabolic pathways for degrading hydrocarbons (Bacosa et al., 2018).

Sediment permeability, which depends on its texture, could explain other discriminant features found in our analyses. The permeable sandy sediments of Lido and Chioggia allow for an advective transport of oxygen as well as particulate and dissolved organic matter in their uppermost layers (Chipman et al., 2012). In contrast, rather impermeable clayey silt sediments, such as those at the innermost lagoon sampling sites, are less efficiently supplied with oxygen and organic matter from the water column (Probandt et al., 2017). This difference influences the structure of the microbial communities present in the two sediment types and, consequently, the discriminant features identified.

Five discriminant features of Palude della Rosa and one of San Giuliano were assigned to the phylum Bacteroidetes, order Bacteroidales and Flavobacteriales. This finding is consistent with the observation that the relative number of cells of this phylum is an order of magnitude greater in cohesive sediments than in highly permeable ones (Probandt et al., 2017). ASV1656, belonging to the genus *Desulfobacter*, has also been found as a discriminant feature in Palude della Rosa, in line with the enrichment of obligate anaerobic bacteria in the less permeable sediments (Chen et al., 2022). At the same time, a higher percentage of sand in Palude della Rosa, compared to Marghera and San Giuliano (average 15.2% with respect to 6.4% and 8.8% respectively), could explain the presence of two ASVs belonging to the Sandaracinaceae family, which is more abundant in sediments with medium permeability (Probandt et al., 2017).

On the other hand, five of the discriminant ASVs identified in Chioggia and Lido belong to some of the most abundant and widespread bacterial families in permeable sediments (Saprospiraceae, Sedimenticolaceae, Halieaceae) (Chen et al., 2022). ASV1621, although belonging to the Flavobacteriaceae, is also a feature found in Lido, in agreement with the presence of the genus *Eudoraea* in medium and highly permeable sediments (Probandt et al., 2017). An intriguing discriminant feature found in the Lido sediments is ASV575. It has been assigned to the genus *Desulfopila*, first described in Japan, where the type species was identified (Suzuki et al., 2007). The species was reported to be a strictly anaerobic and sulphate-reducing bacterium living in estuarine sediments. The detection of this ASV in the permeable Lido sediments seems to corroborate the possibility that Desulfocapsaceae, to which *Desulfopila* belongs, are highly aerotolerant or potentially able to grow aerobically, especially given that MAG-level analysis shows that they encode cytochrome aa_3 oxidases (Chen et al., 2022). ASV1087, which belongs to the genus *RBG-16-58–14* of the family Anaerolineaceae, is another peculiar feature found in the Lido sediments. Indeed, it has been reported that Anaerolineaceae bacteria become dominant in oxygen-depleted sediments (Guo et al., 2022). Conversely, the presence of some genera of the phylum Chloroflexi, including *RBG-16-58-14*, seems to be positively correlated with genes for hydroxylamine and nitrogen dioxide oxidation (Zhang et al., 2021), which would explain the identification of this discriminant feature in sandy sediments.

Other discriminant features found in the lagoon sediment have high potential as indicators of anthropogenic disturbance in marine ecosystems, and were therefore expected in lagoon sediments. In fact, the families Cyclobacteriaceae (ASV5698, in Palude della Rosa), Thiotrichaceae (ASV1145, in Chioggia), Saprospiraceae (with the ASVs already mentioned above) and the genus *Subgroup 23* of the family Thermoanaerobaculaceae (ASV1460 in Marghera and ASV1070 in Palude della Rosa) have been identified among the most significant key indicator variables of disturbed benthic environment (Ramljak et al., 2024).

The presence of two discriminant features, ASV863 in Chioggia and ASV1078 in Lido, assigned respectively to the genus *Anderseniella* and to the genus *Ahrensia*, order Rhizobiales, did not escape our notice. Although the concentrations of dimethylsulphoniopropionate (DMSP) and dimethylsulphide (DMS), as well as their temporal variations, were not measured at the different sites sampled, it is interesting to note that these ASVs could be involved in the DMSP cleavage pathway with the formation of the climate-active gas DMS (Losa et al., 2024; Liu et al., 2016).

Conclusion

This study aimed to provide a general overview of the pelagic and benthic prokaryotic communities in the Venice lagoon's hydrological sub-basins, describing their seasonal changes while also highlighting their long-term stability. Despite the shallow nature of the basin and the frequent sediment resuspension, the two communities exhibited marked differences: only a relatively small number of ASVs were shared between the two domains. Nevertheless, these taxa represent a significant fraction of the total prokaryotic abundance in both water and sediment. Further investigation will be required to determine whether these shared taxa are metabolically active and perform key ecological functions within either environment, particularly through the application of metatranscriptomic approaches.

In general, planktonic prokaryotic communities are rather uniform throughout the lagoon, and are mainly influenced by seasonal variations, with temperature explaining most of the differences in community structure. In contrast, benthic prokaryotic communities are more stable throughout the year, although they differ among sampling sites. Given the different drivers shaping the composition of prokaryotic communities in water and sediment, our data allowed us to identify discriminant ASVs for the seasonal pattern, in water samples, and for the spatial distribution, in surface sediments. In the water, the discriminant ASVs appeared to be related to the cycle of autotrophic organisms, such as phytoplankton and seaweeds. In the sediment, some discriminant ASVs allowed to distinguish between the "less polluted" area (inlet sites and Palude della Rosa) against the "more impacted" zones (San Giuliano - urban discharge, and Marghera - industrial area). Therefore, the sediment microbial community may provide a reliable indication of anthropogenic impacts. However, we unexpectedly found a low abundance of ASVs specialized in hydrocarbon degradation in the community of the industrial area.

The MOSE system became operational in October 2020, but its activations, which included tests on individual inlets, took place on days far apart from those on which the sampling events were carried out. Consequently, the results of this study can be regarded as a reliable reference point for identifying potential changes in the lagoon's prokaryotic communities that may be triggered by the increasingly frequent closures envisaged for the MOSE. Regular monitoring activities should be planned in the next few years, being the MOSE active, to assess possible alterations in benthic prokaryotic community composition resulting from heightened anthropogenic impact within the lagoon. Similarly, future monitoring of water prokaryotic communities will be crucial to determine whether the homogeneous planktonic distribution observed in this study may be disrupted by the hydrodynamic changes associated with frequent MOSE closures, with potential implications for the lagoon ecosystem as a whole.

Data availability statement

The datasets presented in this study can be found in online repositories. The names of the repository/repositories and accession number(s) can be found below: https://www.ncbi.nlm.nih.gov/, PRJNA1096706.

Author contributions

FD: Writing – original draft, Writing – review & editing, Conceptualization, Investigation, Formal Analysis, Validation, Visualization, Methodology, Data curation. GB: Writing – review & editing, Methodology. IG: Writing – original draft, Writing – review & editing, Formal analysis, Visualization. RS: Methodology, Writing – review & editing. EC: Data curation, Methodology, Writing – review & editing. FM: Writing – review & editing. CF: Methodology, Data curation, Writing – review & editing. AV: Methodology, Conceptualization, Supervision, Validation, Investigation, Writing – original draft, Resources, Funding acquisition, Project administration, Writing – review & editing.

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Conflict of interest

The authors declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fmars.2025.1669001/full#supplementary-material

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